

Introduction to Computer Security

Protect information, systems, and people through responsible decisions

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By the end of this course, you will be able to:

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1. **Define security:** describe what computer security protects in plain language

2. **Use the CIA triad:** ask confidentiality, integrity, and availability questions.

3. **Choose a safe response:** Act carefully, ethically, and only with permission

This course teaches protection, not attack

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We will learn

- How security problems happen
- How controls reduce risk
- How to respond safely
- How to think ethically
- How to explain decisions

We will not do

- Attack real systems
- Use real passwords
- Download real malware
- Scan university networks
- Test without permission

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Orientation and Security Basics



Security begins with something valuable

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A

Information

Grades, messages, research, identity data, payment details.

What we are trying to protect

D

Devices + services

Phones, laptops, LMS, email, cloud storage, campus Wi-Fi.

The things we depend on

P

People + reputation

Time, trust, privacy, safety, and the ability to study or work.

Why security ultimately matters

Computer security protects what matters

Protect access • Protect accuracy • Protect use

Computer security is the set of ideas, practices, and controls used to reduce unauthorized access, unwanted change, and harmful disruption.

Your day depends on connected systems

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A single morning may involve:

- unlocking a phone
- opening university email
- joining campus Wi-Fi
- reading an LMS notice
- paying online
- saving work to the cloud



Modern life feels simple because technology hides complexity.
Cybersecurity is partly about understanding those hidden dependencies and what happens when one of them fails or is compromised

Good security avoids two unhelpful extremes

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Too little protection

"Nothing bad has happened, so we are safe."

Result: problems are ignored until an incident causes damage.

Fear without judgment

"Every device and person is an immediate disaster."

Result: useful systems become difficult to use and attention is wasted.

Security needs people, process, and technology

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People
make choices and follow rules.

Process
defines what to do and when.

Technology
blocks, detects, records, or recovers.



Case 1: Meet Mina's university account

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MINA USES ONE ACCOUNT FOR

- LMS lessons
- University emails
- Assignment submission
- Grades
- Shared files

If someone else gets access, what could happen?

Think about:

- information being seen
- information being changed
- Mina being unable to use the account

Case 1: Meet Mina's university account

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1. **8:10 AM - Privacy**
   The attacker reads years of private email and downloads Mina's personal documents.
2. **8:25 AM - Impersonation**
   Messages are sent from Mina's real university email asking classmates and professors for money and files.
3. **9:00 AM - Academic damage**
   Her submitted assignment is deleted or replaced. Important course files disappear.
4. **10:30 AM - Reputation**
   Her professor receives strange messages that appear to come directly from Mina.
5. **11:00 AM - Lockout**
   Mina can no longer access email, the LMS, or her cloud files.
6. **Assignment deadline: 11:59 PM.**

Five words describe a security problem

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ASSET	Something valuable
THREAT	A possible cause of harm
VULNERABILITY	A weakness that could be used or triggered
CONTROL	A safeguard that reduces risk
RISK	The possibility and consequence of harm

What do we value, what could harm it, why could that harm succeed, how serious would the result be, and what can we do about it?

Activity: identify the asset

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A student clicks on a false login link.
An attacker may read university email and reset other accounts.

Which items are assets?

A. The link only **B.** Email, account access, and identity **C.** The attacker only

The CIA Triad

Three questions help us explain almost every basic security goal

The CIA triad asks three protection questions

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C

Confidentiality

Who is allowed to see or access it?

I

Integrity

Is it correct, complete, and trustworthy?

A

Availability

Can authorized people use it when needed?

Confidentiality means authorized access

Confidentiality protects information from being seen or disclosed to people who are not allowed to have it.

School example:
Only the student, authorized instructor, and approved staff should see a private grade.

Key question: Who is allowed to know?



Confidentiality controls access and disclosure

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Everyday controls

- Lock the screen
- Use a strong, unique password
- Turn on multifactor authentication
- Set sharing permissions
- Avoid exposing private data

Confidentiality failures

- Reading another person's email
- Publicly sharing a private file
- Sending data to the wrong person
- Shoulder-surfing a password
- A stolen unlocked device

Integrity keeps information trustworthy

Integrity protects data and systems from improper change or destruction.

School example:
An assignment should arrive exactly as the student submitted it, and a grade should change only through an authorized process.



Key question: Can we trust that it is correct?

Integrity can fail by accident or attack

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Accidental change

- Editing the wrong row
- Uploading an old file
- A software bug changes data
- Storage becomes corrupted
- A user deletes content by mistake

Intentional change

- Altering grades
- Changing payment details
- Modifying a message
- Planting false information
- Deleting evidence

Integrity controls protect trustworthy data

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- Access control limits who can edit.
- Validation checks whether data follows expected rules.
- Logs and version history record what changed.
- Hashes can help detect whether a file changed.
- Backups help restore a known good copy.

Availability means reliable access when needed

Availability keeps systems and information usable for authorized users at the required time.

School example:
The LMS must work when students need lessons, submit assignments, or check announcements.

Key question: Can authorized users use it now?



Availability failures have many causes

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Intentional causes

- Denial-of-service traffic
- Ransomware
- Deliberate deletion
- Account lockout abuse
- Sabotage

Non-malicious causes

- Power failure
- Hardware failure
- Bad software update
- Network congestion
- Fire, flood, or human error

Availability controls support recovery

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- Backups preserve recoverable copies.
- Redundancy provides another component or path.
- Maintenance reduces preventable failures.
- Monitoring detects trouble early.
- Recovery plans explain how to restore service.

CIA questions about the LMS

C	**Who may see each student's grade?**	Only authorized people
I	**Can assignments or grades change improperly?**	Changes are controlled and traceable
A	**Can students log in and submit on time?**	The service remains usable or recovers

Security goals sometimes require balance

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More protection can add effort

Example: multifactor authentication improves confidentiality but adds a login step.

The answer is not to remove protection; it is to make protection usable.

One control can support several goals

Example: a tested backup supports availability and may restore integrity after corruption.

Good designs consider the full system.

Activity: which CIA goal is most affected?

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1. A student's grades are posted publicly. **C I A**

2. Someone changes a submitted assignment. **C I A**

3. The LMS is offline before a deadline. **C I A**

4. A shared file is corrupted and cannot open. **C I A**

Explain CIA without using the letters

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If you can explain it simply, you understand it.

Complete these sentences:

"Confidentiality matters when…"

"Integrity matters when…"

"Availability matters when…"

Everyday Security and Ethics

Most security decisions happen before an alarm

- A message asks you to act quickly.
- A device is left unlocked.
- A file-sharing setting looks too open.
- A friend asks for your account.
- A website requests unexpected information.

The first decision is usually human: continue, pause, verify, or report?

Pause - check - report is a safe first response

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PAUSE

1

Do not click, reply, install, or share.

CHECK

2

Verify the sender, request, address, and context.

REPORT

3

Use the official channel to report.

Urgency is designed to rush the decision

The message says your account will close soon. It provides a link and asks you to sign in immediately.

Warning signs may include:

- unexpected urgency
- an unfamiliar sender
- a mismatched address
- a request for credentials



Pressure is a signal to slow down.

Safe response: stop and verify another way

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1. Do not click the message link.
2. Do not enter a password or authentication code.
3. Open the official university site or app yourself.
4. Contact the help desk using a known address or number.
5. Report the suspicious message and follow guidance.

Permission defines ethical security practice

Authorized practice has a clear owner, scope, time, method, and purpose.

Without explicit permission, even a technically simple test may violate rules, privacy, law, or trust.

When scope is unclear: stop and ask.



Permission changes the ethical status

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Authorized

A student scans an instructor-provided lab machine during the scheduled exercise, using the approved steps, then submits findings privately.

Unauthorized

A student scans a university server or public website because it is reachable, then shares discovered information with friends.

Defense in depth limits one mistake

A safer system uses several layers:

- attentive users
- strong authentication
- device protection
- network controls
- monitoring
- backups and recovery



No single layer is perfect.

Strong security expects that individual controls can fail.

Scenario activity: the urgent LMS message

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At 9:20 p.m., Mina receives a message: "Your university account will be disabled tonight. Confirm immediately." The sender address looks unusual. The link opens a login page that resembles the LMS.

1. What assets are at risk?
2. Which CIA goals could be affected?
3. What should Mina do now?
4. What must Mina avoid doing?

Q & A 1-4 : check your understanding

1. What does computer security protect?

2. What do C, I, and A stand for?

3. **Which CIA goal is most affected when a grade is changed without permission?**

4. Why is explicit permission essential in security practice?

Answers

1. Systems, information, services, people, and other valuable assets.
2. Confidentiality, integrity, and availability.
3. Integrity, because the information was changed improperly.
4. Permission defines ownership, scope, and allowed actions; without it, testing can become misuse.

Q & A 5-8 : check your understanding

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5. Give one control that supports confidentiality.

6. Why can an accidental error still be a security problem?

7. **What should you do first when an urgent message looks suspicious?**

8. Why do secure systems use several layers of defense?

Answers

5. Examples: screen lock, unique password, MFA, or correct sharing permissions.

6. An error can make data incorrect, expose information, or stop a service even without an attacker.

7. Pause, do not click, reply, install, or share. Then verify through a trusted route.

8. Any single control can fail; complementary layers reduce the chance that one mistake causes major harm.



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Thank you!